

Genetic Analysis of Uveal Melanoma in 658 Patients Using the Cancer Genome Atlas Classification of Uveal Melanoma as A, B, C, and D

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Purpose: The Cancer Genome Atlas (TCGA) classification has been validated for uveal melanoma (UM) prognostication. We applied TCGA classification to UM biopsied using fine-needle aspiration biopsy (FNAB) to determine the predictability for metastasis and death.

Design: Retrospective cohort study.

Participants: Patients with UM treated with plaque radiotherapy at Wills Eye Hospital, Philadelphia, Pennsylvania, from October 1, 2008, through December 31, 2018, who completed genetic analysis of chromosomes 3 and 8 after FNAB.

Methods: Tumors were classified as A, B, C, or D and were compared using the chi-square test, Fisher exact test, analysis of variance, and Kaplan-Meier analysis.

Main Outcome Measures: Metastasis and death.

Results: Six hundred fifty-eight UM patients were categorized accordingly as TCGA class A (n = 342 [52%]), B (n = 91 [14%]), C (n = 118 [18%]), and D (n = 107 [16%]). More advanced tumor classification revealed older mean patient age (56 vs. 53 vs. 60 vs. 63 years, respectively; $P < 0.001$), worse presenting visual acuity (20/20–20/50: 81% vs. 67% vs. 71% vs. 66%, respectively; $P < 0.001$), greater distance from the optic disc (3.5 vs. 4.9 vs. 5.7 vs. 5.3 mm, respectively; $P < 0.001$), larger tumor basal diameter (10.3 vs. 12.9 vs. 13.9 vs. 15.3 mm, respectively; $P < 0.001$), and greater tumor thickness (4.3 vs. 6.1 vs. 6.6 vs. 7.5 mm, respectively; $P < 0.001$). After mean follow-up (47.6 vs. 47.6 vs. 42.9 vs. 28.7 months, respectively; $P < 0.001$), more advanced TCGA class was associated with increased risk of metastasis (3% vs. 10% vs. 25% vs. 41%, respectively; $P < 0.001$) and death (1% vs. 0% vs. 3% vs. 9%, respectively; $P < 0.001$). Compared with class A, the 5-year hazard ratio for metastasis increased at 4.1 (B vs. A; $P = 0.01$), 10.1 (C vs. A; $P < 0.001$), and 30.0 (D vs. A; $P < 0.001$). The 5-year hazard ratio for death increased at 3.1 (C vs. A; $P = 0.11$) and 13.7 (D vs. A; $P < 0.001$) with no deaths in class B.

Conclusions: Grouping of UM using TCGA classification predicts the risk of melanoma-related metastasis and death. *Ophthalmology* 2019;126:1445-1453 © 2019 by the American Academy of Ophthalmology

Uveal melanoma (UM) is the most common primary intraocular malignancy of adulthood, with numerous documented clinical, histopathologic, and genetic prognosticators.^{1,2} Despite refinement in the management of UM over the past decade, the mortality rate has remained disturbingly high.³ Understanding that nearly 50% of affected patients eventually will demonstrate metastasis highlights the importance of early detection at a point when the tumor is small and the genetic profile is more favorable.^{2,4} Such patients potentially could benefit from early intervention to reduce tumor de-differentiation into a more malignant process.⁵

Genetic testing using fine-needle aspiration biopsy (FNAB) has earned a fundamental role in modern management of UM.⁶ This tumor analysis has evolved over the past 2 decades from single chromosome 3 analysis to multiple gene

analyses of chromosomes 1, 3, 6, and 8, gene expression profiling, or a combination thereof.^{2,7–10} Most of the available studies in the literature have proven that loss of 1 chromosome 3 and addition of copies of 8q in UM cells are associated with an increased risk of metastasis.¹¹ Shields et al¹² recently determined the personalized risk of UM-related metastasis based on all possible arrays of chromosomes 3 (p and q arms), 6 (p and q arms), and 8 (p and q arms) using Kaplan-Meier analysis, revealing 5-year estimates of metastasis ranging from 4% for those with chromosomes 3, 6, and 8 disomy up to 39% for those with chromosome 3 complete monosomy, chromosome 6 disomy, and chromosome 8q gain. However, correlating and understanding the details of each chromosomal profile can be tedious.

The Cancer Genome Atlas (TCGA) is a project started in 2005 and supervised by the National Cancer Institute's